

UPSTAGE DOCUMENT AGENT

From notice analysis to eligibility — Moabora

Drop any announcement. The agent reads the brief, cross-checks your profile, and answers: **can you apply?**

JunctionX Korea 2026

Upstage Track

Team Aquarium

01 UNDERSTAND

PDF · HWP · Poster Ingest

02 JUDGE

Automated Eligibility Verdict

03 ACT

Solar Fit Score & Checklist

Notices are everywhere. Eligibility is not.

Students miss vital opportunities not from lack of listings, but from the exhausting cost of manual reading.

PAIN 01

Fragmented Document Formats

- Scattered across PDF guidelines, HWP attachments, image posters, and bulletin HTML.
- Unstructured binary formats that search engines and keyword filters cannot parse.

Bottleneck: Format Incompatibility

PAIN 02

Buried Qualification Clauses

- Crucial constraints ("under 29", "no 4th years", "IT majors only") hidden deep inside multi-page briefs.
- Students must inspect every page just to discover they are disqualified.

Bottleneck: Information Opacity

PAIN 03

Exhausting Reading Overhead

- Cross-checking academic status, age, and major takes 15–30 minutes per notice.
- Evaluating 20+ opportunities requires hours of repetitive, manual toil.

Bottleneck: Time & Cognitive Burden

PAIN 04

Fragmented Channels & Missed Deadlines

- Notices spread across campus portals, Wevity, and ContestKorea without sync.
- Crucial application deadlines pass unnoticed with zero personalized tracking.

Bottleneck: Opportunity Loss

The bottleneck is reading, not listing

Existing aggregators stop at collecting links. The real friction begins after you click.

EXISTING NOTICE BOARDS

List Aggregators (Wevity, ContestKorea)

DOCUMENT ACCESS

Hands you raw attachments. You must download and read HWP/PDFs yourself.

ELIGIBILITY CHECK

100% manual. User spends 15–30 minutes deciphering complex guideline clauses.

RECOMMENDATION

Generic category tags with zero awareness of your actual skills or profile.

Outcome: Manual Workload Pushed to Student

MOABORA DOCUMENT AGENT

Autonomous Document Agent

DOCUMENT ACCESS

Upstage AI reads & structures HWP, PDF, and poster images automatically.

ELIGIBILITY CHECK

Instant 6-rule matching (Eligible Review Ineligible) with explicit reasons.

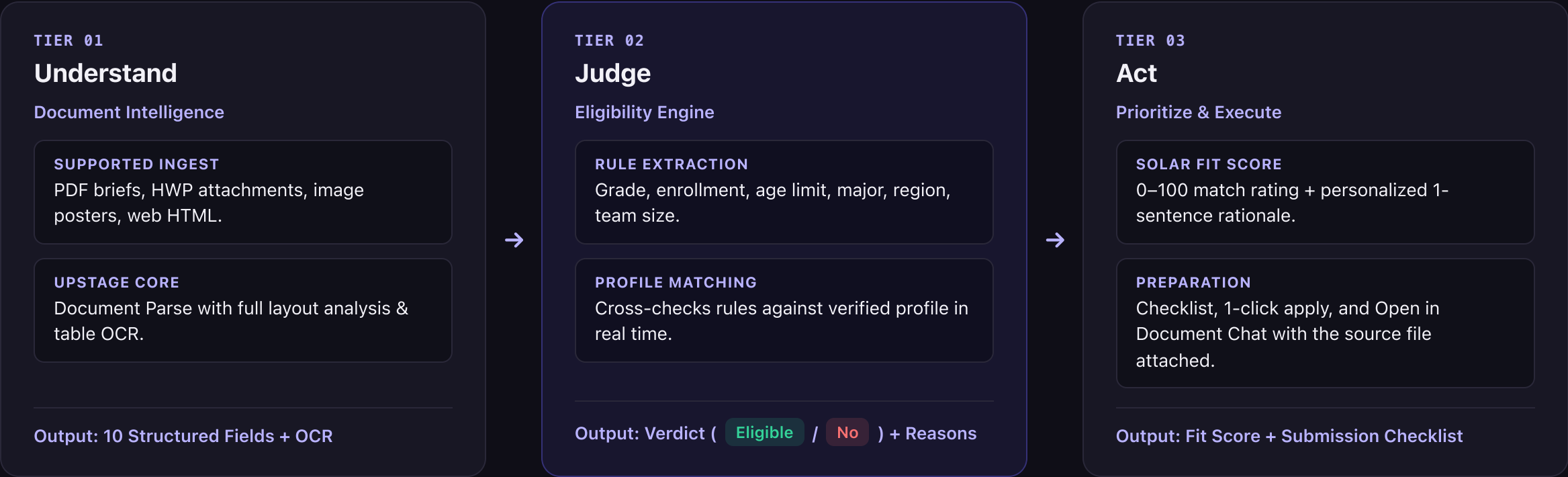
RECOMMENDATION

Solar Pro 4 calculates personalized fit score (0–100) + tailored rationale.

Outcome: Instant Verdict & Submission Checklist

Understand → Judge → Act

A three-tier pipeline transforming raw announcement files into verified application actions.



From zero to application in 4 steps

A frictionless 2-minute workflow from profile setup to finalized application decision.

STEP 01 · 1-MINUTE ONBOARDING

Profile Creation

- Certificate (UIE), GitHub/blog link, manual fields, or a short bio (Solar).
- Merges university, year, status, skills, and interests from any path.

Deliverable: Verified Student Profile

STEP 02 · MULTI-CHANNEL INGEST

Notice Collection

- Drag & drop PDF / HWP files, paste announcement URLs, or 1-click crawler.
- Automated collection from Wevity and ContestKorea with HWP attachment download.

Deliverable: Multi-Modal Ingestion Pipeline

STEP 03 · PERSONALIZED RANKING

Feed & Fit Scoring

- Feed displays instant eligibility status badges (**Eligible** **Review** **Ineligible**).
- Solar Pro 4 calculates tailored fit score (0–100) with personalized match reasons.

Deliverable: Intelligent Ranked Opportunity Feed

STEP 04 · PREPARATION & ACTION

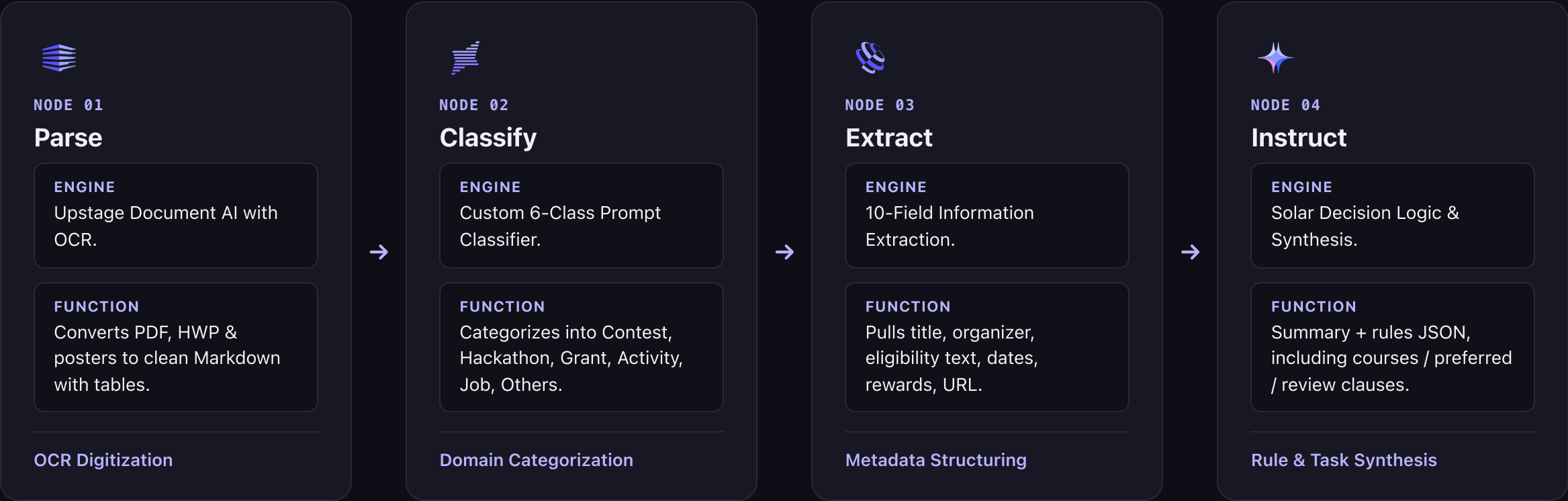
Execution & Doc Chat

- Review items become a confirm-checklist when the verdict is Needs review.
- Open in Document Chat from the notice — source file attached, prompt pre-filled.

Deliverable: Verified Application Submission

Every notice flows through one 4-node agent

All ingest paths (upload, link, crawler) execute the Studio Agent via REST API with zero bypass.



All four product lines in production

Comprehensive integration across Upstage's Document AI and LLM ecosystem.



Studio Agents

PRIMARY ROLE
Core 4-Node Notice Processing Pipeline (`agt\_...`).

ARCHITECTURE
Accumulating snapshot map ensures full node visibility.

REST /v2/responses



Document Parse

PRIMARY ROLE
High-Precision OCR & Layout Analysis.

INTEGRATION
Powers `parse\_document` tool in Document Chat.

/v1/document-digitization



Information Extract

PRIMARY ROLE
Zero-Shot Schema-Driven Document Extraction.

INTEGRATION
Certificate parsing & `extract\_information` chat tool.

/v1/information-extraction



Solar Pro 4

PRIMARY ROLE
Decision Intelligence & Multi-Tool Orchestration.

INTEGRATION
Fit scores, link/note profile extract, streamed tool-calling.

Direct API (OpenAI-Compatible)

Service-grade reliability & transparency

Engineered to handle real-world document edge cases and provide full visibility into AI execution.

ROBUSTNESS

JSON Sanitization

Pre-processes double-encoded Studio Instruct outputs and strips citation markers ([+1]) seamlessly.

Parser Resilience

STATE MANAGEMENT

Snapshot Accumulation

Maintains an accumulating map across Studio polling snapshots so intermediate node outputs are preserved.

Full Node Visibility

TRANSPARENCY

Live NDJSON Streams

Streams pipeline stages, intermediate artifacts, and Solar reasoning tokens in real time to the client.

Streamed Reasoning

RELIABILITY

Resumable Sessions

Server-side workflow sessions survive browser refreshes, preventing duplicate execution and billing.

Refresh-Safe Architecture

OPTIMIZATION

Dual-Layer Caching

SHA-256 document parse cache + profile/notice hash-keyed Solar recommendation cache avoids redundant calls.

Cost & Latency Control

ACCURACY

Deterministic Matcher

Evaluates 6 criteria (grade, status, age, major, region, team) with clear per-rule reason breakdown.

Explainable AI Verdict

3-day hackathon execution & future milestones

Prioritized a rock-solid end-to-end loop while laying an extensible foundation for production scale.

SHIPPED MVP

Delivered in 3 Days

- 3 Ingest paths (Files, URLs, 2-site Crawlers).
- 4-node Studio Agent with full error handling.
- 3 profile paths (link, certificate, manual) + Solar note fill.
- Eligibility engine + Review checklist for ambiguous clauses.
- Live NDJSON streams & Doc Chat (notice handoff + 3 tools).
- Supabase Cloud storage + Docker deployment.

100% Production Ready

CONSCIOUSLY SCOPED OUT

Deferred Complexity

- **Multi-Tenant Auth:** Single-profile demonstration model.
- **Push Notifications:** Excluded background email/push daemons.
- **Scheduled Crawlers:** Manual trigger-based crawling only.

Avoided Low-Demo Bloat

NEXT MILESTONES

Production Roadmap

- **Multi-User Auth:** Supabase Auth + RLS (schema is RLS-ready).
- **Proactive Alerts:** D-3 deadline notifications via Telegram/Discord.
- **Application Drafts:** Solar-assisted personal statement generator.
- **Team Matching:** Connect students applying to the same contest.

Clear Scaling Path

Four bets we actually shipped

Depth of the stack, a complete loop, a sharper problem, and honest scoping — not a demo shell.

Upstage Technology

- Custom 4-node Studio Agent (`agt\_...`) via REST v2 APIs without bypass.
- Production deployment of all 4 Upstage product lines.
- Robust handling of double-encoded JSON and live streaming reasoning.

Max Technical Complexity

Service Completeness

- End-to-end loop: Landing → Profile → Crawl/Ingest → Feed → Detail → Chat.
- Live data from Wevity & ContestKorea with real HWP attachments.
- Resumable workflow sessions that survive browser refresh.

Full Production Loop

Idea Creativity

- Shifting paradigm from "Listing links" to "Delivering application verdicts".
- Reads HWP attachments and posters that traditional aggregators skip.
- Transparent UX displaying live agent thought process and rule proofs.

Novel Problem Framing

Product Planning

- Clear problem focus: Relieving student document fatigue.
- Thoughtful MVP scoping with explicit trade-offs and roadmap.
- Fully documented architecture, data models, and RLS-ready schema.

Execution Rigor

LIVE DEMO

Try Moabora live in production

Fully deployed autonomous document agent running on Supabase Cloud & Upstage Studio Agent APIs.

[https:// moabora.sharp0802.com](https://moabora.sharp0802.com)